KEDGE Business School EBP1 Quantitative Methods 1

# Worksheet 5 — Time series

**Fully worked solutions**

Badr ABOUFARASSE, PhD — 2026–2027

## 1 Core applications

Master the essential definitions and calculations from the course.

*Core application — 10 min*

### 1.1 Exercise 1 – Components and models

1. Match long-run trend, recession, recurring Q4 peak and exceptional shock to $T_t$, $C_t$, $S_t$ or $R_t$.
2. Write the full additive and multiplicative models.
3. Explain when multiplicative seasonality is more plausible than additive seasonality.

**Fully worked solution**

$$Y_t = T_t + C_t + S_t + R_t,$$

$$Y_t = T_t \times C_t \times S_t \times R_t.$$

Multiplicative seasonality is more plausible when seasonal amplitude grows with the level of the series.

**Distinguishing additive units from multiplicative factors**

An additive seasonal component is measured in units of Y and has a constant absolute amplitude. A multiplicative seasonal component is a factor (e.g. 1.20 for +20%), so its absolute effect varies with the trend level. A cycle spans longer fluctuations, whereas an irregular shock is not periodic.

*Core application — 20 min*

### 1.2 Exercise 2 – Moving averages and centring

Quarterly demand over two years is

$$80,\ 100,\ 120,\ 140,\ 88,\ 110,\ 132,\ 154.$$

Describe the main pattern, calculate the five 4-quarter moving averages, centre them for periods 3 to 6, and explain why an even-order moving average must be centred.

**Fully worked solution**

$$MA_{2.5} = 110, \quad MA_{3.5} = 112, \quad MA_{4.5} = 114.5, \quad MA_{5.5} = 117.5, \quad MA_{6.5} = 121.$$

$$CMA_3 = 111, \quad CMA_4 = 113.25, \quad CMA_5 = 116, \quad CMA_6 = 119.25.$$

An even-order moving average lies between two periods, so averaging adjacent moving averages centres the estimate on an observed period.

**Moving-average alignment table**

| Window | Sum | MA4 | Midpoint |
|---|---|---|---|
| 1–4 | 440 | 110 | 2.5 |
| 2–5 | 448 | 112 | 3.5 |
| 3–6 | 458 | 114.5 | 4.5 |
| 4–7 | 470 | 117.5 | 5.5 |
| 5–8 | 484 | 121 | 6.5 |

| t | 3 | 4 | 5 | 6 |
|---|---|---|---|---|
| CMA4 | (110+112)/2=111 | (112+114.5)/2=113.25 | (114.5+117.5)/2=116 | (117.5+121)/2=119.25 |

Because order 4 is even, the first moving average is centred between t=2 and t=3. Centring gives values at integer dates t=3 through 6.

## 2 Assessment and exam practice

Practise independently with assessment and final-exam style questions.

*Assessment / exam practice — 25 min*

### 2.1 Exercise 3 – Multiplicative decomposition and forecasting

| Year | Q1 | Q2 | Q3 | Q4 |
|---|---|---|---|---|
| 2005 | 82 | 98 | 120 | 150 |
| 2006 | 91 | 111 | 137 | 171 |
| 2007 | 104 | 126 | 154 | 194 |

Given $\widehat{T}_t = 6.343t + 86.939$ and seasonal coefficients $0.789, 0.903, 1.052, 1.255$, interpret the trend slope, check normalization, forecast the four quarters of 2008, and explain multiplicative deseasonalization.

**Fully worked solution**

The trend slope 6.343 implies a 6.343-unit increase per quarter. The supplied seasonal factors sum to 3.999, approximately 4 (rounding).

| t | Q | $T_t$=6.343t+86.939 | S' | $T_t \times$S' |
|---|---|---|---|---|
| 13 | Q1 | 169.398 | 0.789 | 133.655022 |
| 14 | Q2 | 175.741 | 0.903 | 158.694123 |
| 15 | Q3 | 182.084 | 1.052 | 191.552368 |
| 16 | Q4 | 188.427 | 1.255 | 236.475885 |

$$\widehat{Y}_{13} = 169.398(0.789) = 133.655022, \quad \widehat{Y}_{14} = 175.741(0.903) = 158.694123.$$

$$\widehat{Y}_{15} = 182.084(1.052) = 191.552368, \quad \widehat{Y}_{16} = 188.427(1.255) = 236.475885.$$

To deseasonalise an observation, divide $Y_t$ by the relevant quarterly factor S'. Retain unrounded figures until the final result.

*Assessment / exam practice — 65 min*

### 2.2 Exercise 5 — Full seasonal decomposition — exam style

A firm observes quarterly sales over three years. Use the multiplicative model $Y_t = T_t \times S_t$, without explicit cycle or irregular component. Do not round intermediate values.

| Année / Year | Q1 | Q2 | Q3 | Q4 |
|---|---|---|---|---|
| 2019 | 40 | 52 | 48 | 68 |
| 2020 | 44 | 58 | 53 | 76 |
| 2021 | 50 | 65 | 60 | 84 |

1. Plot the series and calculate all nine 4-quarter moving averages.
2. Centre the moving averages (t=3 to 10). Build the t, CMA4, $t^2$, t×CMA4 table.
3. Estimate trend $T_t = at + b$ by OLS on the eight centred moving averages.
4. Calculate $Y_t/T_t$ for all 12 quarters, the four seasonal means and normalised factors summing to 4.
5. Forecast all four quarters of 2022 (t=13 to 16), first deseasonalised, then seasonalised.

**Fully worked solution**

1. Four-quarter moving averages:

$$MA_{2.5} = (40 + 52 + 48 + 68)/4 = 52, \quad MA_{3.5} = (52 + 48 + 68 + 44)/4 = 53.$$

| Fenêtre / Window | Périodes / Periods | Somme / Sum | MA4 | Centre / Midpoint |
|---|---|---|---|---|
| 1 | 1–4 | 208 | 52 | 2.5 |
| 2 | 2–5 | 212 | 53 | 3.5 |
| 3 | 3–6 | 218 | 54.5 | 4.5 |
| 4 | 4–7 | 223 | 55.75 | 5.5 |
| 5 | 5–8 | 231 | 57.75 | 6.5 |
| 6 | 6–9 | 237 | 59.25 | 7.5 |
| 7 | 7–10 | 244 | 61 | 8.5 |
| 8 | 8–11 | 251 | 62.75 | 9.5 |
| 9 | 9–12 | 259 | 64.75 | 10.5 |

2. Centring: the nine half-period MA4 values produce eight CMA4 values at integer periods 3 to 10.

$$CMA_3 = (52 + 53)/2 = 52.5, \quad CMA_4 = (53 + 54.5)/2 = 53.75.$$

| t | CMA4 | $t^2$ | t×CMA4 |
|---|---|---|---|
| 3 | 52.5 | 9 | 157.5 |
| 4 | 53.75 | 16 | 215 |
| 5 | 55.125 | 25 | 275.625 |
| 6 | 56.75 | 36 | 340.5 |
| 7 | 58.5 | 49 | 409.5 |
| 8 | 60.125 | 64 | 481 |
| 9 | 61.875 | 81 | 556.875 |
| 10 | 63.75 | 100 | 637.5 |
| Σt=52 | 462.375 | 380 | 3073.5 |

3. Trend regression using CMA4 values:

$$\bar{t} = 52/8 = 6.5, \quad \overline{CMA} = 462.375/8 = 57.796875.$$

$$V(t) = 380/8 - 6.5^2 = 5.25, \quad \mathrm{Cov}(t, CMA) = 3073.5/8 - 6.5(57.796875) = 8.5078125.$$

$$a = 8.5078125/5.25 = 1.620535714, \quad b = 57.796875 - a(6.5) = 47.263392857.$$

$$\widehat{T}_t = 1.620535714t + 47.263392857.$$

4. Individual seasonal ratios and normalisation:

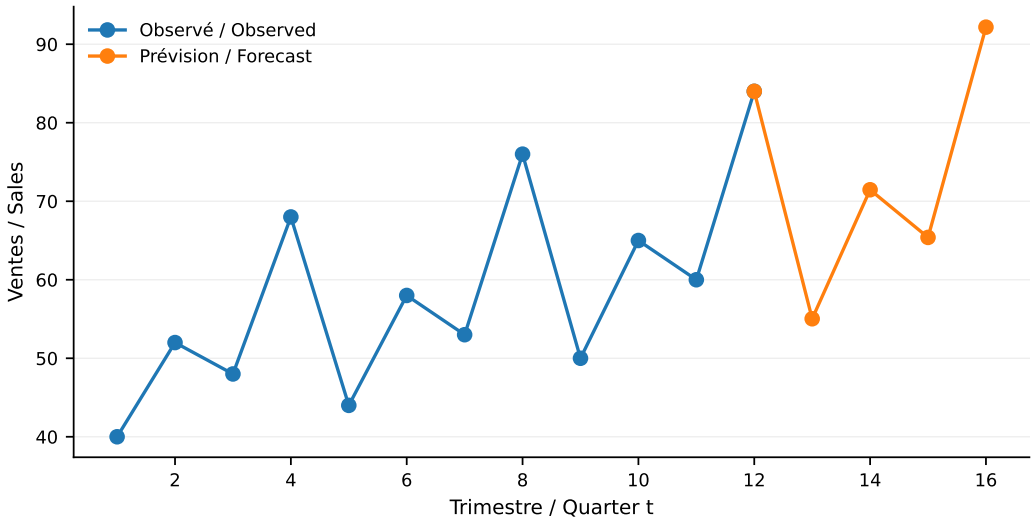
| t | Trimestre / Quarter | Y | $T_t$ | $S_t$=Y/$T_t$ |
|---|---|---|---|---|
| 1 | Q1 | 40 | 48.8839 | 0.81826 |
| 2 | Q2 | 52 | 50.5045 | 1.02961 |
| 3 | Q3 | 48 | 52.125 | 0.92086 |
| 4 | Q4 | 68 | 53.7455 | 1.26522 |
| 5 | Q1 | 44 | 55.3661 | 0.79471 |
| 6 | Q2 | 58 | 56.9866 | 1.01778 |
| 7 | Q3 | 53 | 58.6071 | 0.90433 |
| 8 | Q4 | 76 | 60.2277 | 1.26188 |
| 9 | Q1 | 50 | 61.8482 | 0.80843 |
| 10 | Q2 | 65 | 63.4688 | 1.02413 |
| 11 | Q3 | 60 | 65.0893 | 0.92181 |
| 12 | Q4 | 84 | 66.7098 | 1.25918 |

$$\bar{S}_q = \frac{1}{3}\sum_{j=0}^{2}(Y_{4j+q}/\widehat{T}_{4j+q}), \qquad S'_q = \frac{4\bar{S}_q}{\sum_{k=1}^{4}\bar{S}_k}.$$

| Trimestre / Quarter | Ratios / Ratios | Moyenne / Mean | S' normalisé / Normalised |
|---|---|---|---|
| Q1 | 0.81826 ; 0.79471 ; 0.80843 | 0.807135 | 0.805376 |
| Q2 | 1.02961 ; 1.01778 ; 1.02413 | 1.02384 | 1.021609 |
| Q3 | 0.92086 ; 0.90433 ; 0.92181 | 0.915667 | 0.913671 |
| Q4 | 1.26522 ; 1.26188 ; 1.25918 | 1.262095 | 1.259344 |
| TOTAL | — | 4.008737 | 4 |

5. Forecast: $T_t$ is the deseasonalised forecast; multiply by the relevant quarterly factor S' for the final forecast.

| t | Trimestre / Quarter | $T_t$ | S' | Prévision / Forecast |
|---|---|---|---|---|
| 13 | Q1 | 68.3304 | 0.805376 | 55.0316 |
| 14 | Q2 | 69.9509 | 1.021609 | 71.4624 |
| 15 | Q3 | 71.5714 | 0.913671 | 65.3927 |
| 16 | Q4 | 73.192 | 1.259344 | 92.1739 |




*Ventes observées et prévisions 2022 / Observed and 2022 forecasts*

Check: the four normalised seasonal factors sum to 4. The 2022 forecasts are approximately 55.03, 71.46, 65.39 and 92.17.

## 3 Optional advanced practice

Explore additional interpretations and professional applications.

*Advanced practice — 15 min*

### 3.1 Exercise 4 – Forecast evaluation

For this exercise, suppose actual 2008 values were $136, 162, 188, 245$. Using the Exercise 3 forecasts, calculate forecast errors, MAE and RMSE, and explain the value of a separate validation period and a naive benchmark.

**Fully worked solution**

Forecasts are exactly those computed in exercise 3, without substituting rounded figures from another source.

| Q | Y | Prévision / Forecast | e=Y−$\widehat{Y}$ | \|e\| | $e^2$ |
|---|---|---|---|---|---|
| 1 | 136 | 133.655022 | 2.344978 | 2.344978 | 5.49892 |
| 2 | 162 | 158.694123 | 3.305877 | 3.305877 | 10.92882 |
| 3 | 188 | 191.552368 | -3.552368 | 3.552368 | 12.61932 |
| 4 | 245 | 236.475885 | 8.524115 | 8.524115 | 72.66054 |

$$MAE = (2.344978 + 3.305877 + 3.552368 + 8.524115)/4 = 4.431835.$$

$$RMSE = \sqrt{(2.344978^2 + 3.305877^2 + (-3.552368)^2 + 8.524115^2)/4} = 5.042509.$$

$$ME = (2.344978 + 3.305877 - 3.552368 + 8.524115)/4 = 2.655651.$$

The model underpredicts on average. Validation should be out of sample; a naive forecast is a useful benchmark.

*Advanced practice — 20 min*

### 3.2 Extra 1 — Forecast validation and structural break

Actual monthly demand is 100, 104, 108, 150, 154 and 158. A trend model fitted on the first three months predicts 100, 104, 108, 112, 116 and 120.

1. Calculate forecast errors for months 4–6.
2. Calculate MAE and mean error over the validation period.
3. Diagnose the model.
4. Propose a responsible forecasting response.

**Method hint:**

Use error = actual − forecast. A persistently positive error signals systematic underforecasting.

**Fully worked solution**

**Errors.** Month 4: 150−112 = 38; month 5: 154−116 = 38; month 6: 158−120 = 38.

**Validation metrics.** MAE = (38+38+38)/3 = **38**. Mean error is also **+38**, showing systematic underforecasting rather than random noise.

**Diagnosis.** The level jumps between months 3 and 4 while the subsequent slope remains close to four units per month. This is consistent with a level structural break. A model trained only before the break is obsolete.

**Response.** Investigate the business event causing the jump, re-estimate the level using post-break data, compare with a naive benchmark, retain a validation window, and issue prediction intervals. Do not simply average pre- and post-break observations without explaining the regime change.